

MUHAMMED ALTHAF K

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PROFESSIONAL SUMMARY

B.Tech graduate in Computer Science (Data Science and Machine Learning) with hands-on experience in end-to-end machine learning pipelines, exploratory data analysis, dashboard development, and statistical modeling. Proficient in Python, SQL, Power BI, and Excel with a strong foundation in feature engineering, predictive modeling, and data visualization. Experienced in translating complex data findings into clear, actionable insights for business stakeholders.

TECHNICAL SKILLS

Languages: Python, SQL

Data Analysis & Visualization: Power BI, Tableau, Excel (Advanced), DAX, Power Query

Libraries: Pandas, NumPy, Matplotlib, Seaborn, Streamlit, SciPy, Joblib

Core Skills: Data Cleaning, EDA, Data Wrangling, Statistical Analysis, Feature Engineering, Dashboard Development

Platforms: Jupyter Notebook, Google Colab, GitHub, Oracle Cloud Infrastructure

PROJECTS

Silent Burnout AI - Student Disengagement Detection | Python, Scikit-learn, Streamlit

May 2026

- Built an end-to-end machine learning pipeline using the OULAD dataset; engineered 10 behavioral and academic features including LMS click patterns, active days, engagement variability, and assessment score trends to classify students into Low, Moderate, or High burnout risk categories.
- Benchmarked 6 ML models including Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and Gradient Boosting; selected Gradient Boosting as the best model; deployed a live Streamlit dashboard at silent-burnout.streamlit.app with risk predictions, confidence scores, and engagement radar profiles.

Shopify Customer and Sales Analysis Dashboard | Power BI, DAX, Excel, Power Query

Aug 2025

- Designed a 2-page interactive Power BI dashboard analyzing 7,431 Shopify transactions and tracking 9 KPIs including Net Sales, Average Order Value, Repeat Rate, Customer Lifetime Value, and Purchase Frequency with regional maps and product-type breakdowns.
- Automated data transformations using Power Query and DAX measures; delivered stakeholder-ready visuals simplifying complex retail trends for non-technical business teams.

Laptop Price Prediction | Python, Scikit-learn, Random Forest, Pandas

May 2025

- Processed 1,300+ records with 8 engineered attributes; applied log transformation on the target variable; compared Linear Regression, KNN, and Random Forest models - Random Forest achieved R-squared of 0.887 and MAE of 0.159.

INTERSHIP AND TRAINING

Traffic Forecasting Mentorship | MentorMind x Uber x upGrad

Aug 2025 - Oct 2025

- Built a predictive machine learning model to forecast hourly traffic volumes at road junctions by integrating multi-source data including historical traffic records, weather data (temperature, precipitation, wind speed), and event data into a unified pipeline.
- Engineered time-based features including hour of day, day of week, lag values, and binary event indicators; performed peak hour analysis using congestion metrics, moving averages, and heatmap visualizations.
- Trained and compared ARIMA, LSTM, and Gradient Boosting models; evaluated performance using MAE, RMSE, and R-squared with k-fold and time-based cross-validation; optimized hyperparameters using Grid Search and Random Search.

Complete Machine Learning and Data Science Program | GeeksForGeeks

Jun 2024 - Jul 2024

- Completed a 90-hour applied program covering exploratory data analysis (EDA), predictive modeling, and visualization using Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

CERTIFICATIONS

- Oracle Cloud Infrastructure - Certified Data Science Professional | Oracle (Oct 2025)
- Complete Machine Learning and Data Science | GeeksForGeeks (Jul 2024)
- Become a Data Scientist | LinkedIn Learning (Mar 2023)

EDUCATION

B.Tech - CSE (Data Science with Machine Learning) | Lovely Professional University, Punjab

May 2026

Class XII - Science | Seethi Sahib Higher Secondary School, Kerala | 95%

Mar 2022

Class X | Seethi Sahib High School, Kerala | 95%

Mar 2020